Deep Reinforcement Learning for Risk-Adjusted Cryptocurrency Portfolio Management

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Abstract

Deep reinforcement learning agents trained for portfolio management frequently achieve strong simulated returns while accumulating transaction costs that erode real-world profitability. An initial investigation confirmed this pattern: a Proximal Policy Optimization (PPO) agent applied to cryptocurrency assets rebalanced without restraint, accumulating transaction fees that routinely exceeded the agent's actual alpha. This work addresses the problem through a three-component framework. First, a price-drift-adjusted no-trade zone suppresses rebalancing unless the deviation between target and naturally drifted weights exceeds a threshold of 5%. Second, a cash allocation slot is incorporated as a learnable defensive position, allowing the agent to reduce market exposure without any hard-coded exit rule. Third, a quadratic drawdown penalty escalates as losses deepen, creating a strong incentive to seek capital preservation under adverse conditions. Evaluated on BTC, ETH, and BNB from 2020 to 2024 using walk-forward validation, the proposed system reduced annual transaction costs below 0.6% while maintaining competitive risk-adjusted performance. Systematic experiments across three reward formulations, two network architectures, and three random seeds confirm the reproducibility of these findings.

Keywords: deep reinforcement learning, portfolio management, cryptocurrency, Proximal Policy Optimization, transaction costs, no-trade zone

1 Introduction

Cryptocurrency markets have grown from a niche curiosity into a multi-trillion dollar asset class over the past decade. This expansion has attracted substantial research attention in automated trading, where the high volatility and continuous 24/7 operation of crypto markets present challenges that are qualitatively different from traditional equity environments. BTC can lose 15% in a single session. Correlated drawdowns frequently affect multiple assets simultaneously. Deep reinforcement learning offers a principled approach to this setting: agents learn allocation strategies directly from market data without requiring explicit price predictions, and multiple prior studies have demonstrated that RL agents can outperform naive benchmarks on simulated crypto markets [1, 2].

A critical limitation, however, has received insufficient attention in prior work: transaction costs. Binance charges 0.1% per trade. Cryptocurrency markets operate continuously (24/7), and an agent that rebalances without restraint can accumulate fees that far exceed any realistic alpha. Preliminary experiments in this study confirmed this pattern directly: an unconstrained PPO agent generated substantial gross returns but net returns were severely eroded by accumulated transaction costs.

The root cause is architectural. Conventional RL environments for portfolio management compare target weights against the *previous day's* weights. Since prices move continuously, this creates artificial deviations that trigger unnecessary trades. A concrete example illustrates the issue: a portfolio holds 26% BTC. Overnight, BTC climbs 1%, and natural price drift pushes BTC's weight to 26.2%. If the agent's new target is 26.3%, a naive system executes a trade to close a 0.1% gap, paying a fee that exceeds any conceivable benefit. Repeated daily across a full year, the losses are substantial.

This study proposes three complementary interventions. First, a **price-drift-adjusted no-trade zone** measures the deviation between target weights and naturally drifted weights rather than previous-day weights. Rebalancing occurs only when this deviation exceeds 5%, eliminating fee-negative trades while preserving intentional repositioning. Second, **cash as a learnable asset** provides an allocation refuge without any hard-coded market-exit rule; the agent discovers through reward shaping when capital preservation is preferable to market exposure. Third, a **quadratic drawdown penalty** in the reward function creates progressively stronger incentives to reduce exposure as losses deepen, mirroring the mathematical asymmetry of drawdown recovery documented in classical portfolio theory [3].

The specific contributions of this work are:

1. A **price-drift adjusted no-trade zone** that substantially reduces annual transaction costs, to below 0.6% in experiments, without sacrificing risk-adjusted performance.
2. **Cash as a learned refuge.** The agent autonomously shifts capital into cash under adverse conditions, without any hand-coded exit rule, simply by incorporating cash as a fourth allocation target in the action space.
3. A **four-component reward function**: log return for growth, downside volatility penalty for risk, turnover penalty to discourage excessive rebalancing, and a

quadratic drawdown term that provides escalating incentives for capital preservation.

4. **Reproducible experiments** across three reward formulations, two network architectures (MLP and LSTM), two training windows, and three random seeds, with both mean and standard deviation reported.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the dataset and preprocessing procedure. Section 4 presents the full methodology. Section 5 defines the experimental protocol. Section 6 reports and interprets results. Section 7 presents a live out-of-sample evaluation covering March 2025 through March 2026. Section 8 identifies limitations. Section 9 concludes.

2 Related Work

2.1 DRL for Portfolio Management

The EIIE framework by Jiang et al. [1] was among the first systems to apply neural networks to cryptocurrency asset allocation. Their architecture combined CNN and RNN layers with a Portfolio-Vector Memory, and demonstrated that agents could learn effective trading strategies directly from raw price data. Transaction costs in that work were modeled as a single fixed rate applied regardless of trade frequency. For agents executing hundreds of trades per year, this simplification significantly underestimates realized costs.

For equity markets, Yang et al. [4] proposed an ensemble approach combining PPO, A2C, and DDPG, switching between algorithms based on recent Sharpe ratio performance. Liu et al. [5] extended this direction into FinRL, an open-source library providing standardized RL environments for quantitative finance research. Both contributions focus primarily on equities; direct application to cryptocurrency markets presents additional challenges including substantially higher volatility, continuous trading hours, and periodic liquidity crises in smaller tokens.

2.2 Risk-Aware Reward Design

A reward function that maximizes only raw return produces agents that concentrate positions in recently performing assets, creating portfolios vulnerable to sudden reversals. Incorporating explicit risk penalties into the reward signal is a well-established mitigation strategy. Choudhary et al. [6] systematically compared multiple risk-adjusted reward formulations and found that explicit risk penalties reliably reduce peak drawdowns, at modest cost to raw returns. Park et al. [7] embedded Conditional Value-at-Risk (CVaR) directly into the policy gradient step. The approach in this study attaches three penalty terms to the reward (downside volatility, turnover, and quadratic drawdown) and allows the agent to balance them through experience. This design is both simpler to implement and more interpretable than CVaR embedding, since each penalty term is directly observable and attributable.

2.3 Transaction Cost Management

The divergence between backtest and live performance in RL trading systems is frequently attributable to inadequate cost modeling. Betancourt and Chen [2] developed a DRL system with explicit exchange fee optimization. Sadighian [8] demonstrated that backtest results without realistic cost models systematically overfit to profits that would not materialize in practice. The approach proposed in this study differs in emphasis: rather than optimizing execution given that a trade occurs, the system first determines whether a trade is cost-justified. The no-trade zone functions as a gating mechanism; a rebalancing action that does not clear the threshold is suppressed entirely. Betancourt and Chen [2] reached a similar empirical finding: once fees are properly modeled, small weight adjustments contribute negligible net returns.

2.4 Drawdown Control

Grossman and Zhou [3] established the theoretical optimality of reducing market exposure as drawdown grows under a hard maximum drawdown constraint. The intuition is reinforced by the mathematical asymmetry of recovery: a 10% loss requires an 11.1% gain to recover; a 40% loss requires a 66.7% gain. The quadratic drawdown penalty in this study incorporates this asymmetry as a reward signal, creating progressively stronger incentives to reduce risk exposure as losses deepen.

3 Data

3.1 Dataset

Daily OHLCV data (Open, High, Low, Close, Volume) for Bitcoin (BTC), Ethereum (ETH), and Binance Coin (BNB) were collected via the Yahoo Finance API, covering January 1, 2020 through December 31, 2024, producing 1,826 aligned trading days after cleaning. These three assets were selected on the basis of market capitalization and order-book depth on Binance.

3.2 Features

Six technical indicators per asset, all derived from the closing price series:

1. **Close** (P_t): the base price series.
2. **7-day moving average** ($\mathrm{MA7}_t$): short-term trend.
3. **21-day moving average** ($\mathrm{MA21}_t$): medium-term trend. The MA7/MA21 crossover is a widely observed directional signal.
4. **RSI, 14-day**: a momentum oscillator bounded to $[0, 100]$. Values above 70 indicate overbought conditions; values below 30 indicate oversold conditions.
5. **MACD** ($\mathrm{EMA}_{12} - \mathrm{EMA}_{26}$): positive when short-term momentum exceeds long-term momentum.
6. **14-day realized volatility**: standard deviation of daily log returns over the prior two weeks, providing a direct signal of recent market uncertainty.

3.3 Normalization

A rolling z-score with a 60-day lookback is applied to all features:

$$z_t = \frac{x_t - \hat{\mu}_t}{\hat{\sigma}_t + \epsilon}, \quad \hat{\mu}_t = \frac{1}{60}\sum_{k=1}^{60} x_{t-k}, \quad \hat{\sigma}_t = \sqrt{\frac{1}{59}\sum_{k=1}^{60}(x_{t-k} - \hat{\mu}_t)^2} \tag{1}$$

The rolling window is essential to prevent look-ahead bias. A global z-score computed over the entire dataset would embed future statistical information into early training periods. With a rolling window, all normalization statistics are derived exclusively from past observations.

3.4 Data Cleaning

Days on which any asset moved more than 50% in a single session were flagged for review. One such event was identified: BNB on February 19, 2021, associated with a Binance Smart Chain deployment event rather than genuine market activity. This observation was removed, and the three time series were re-aligned by retaining only dates present across all three assets.

3.5 Walk-Forward Validation

Walk-forward validation replicates the sequential nature of real deployment: train on historical data, evaluate on the subsequent period, then advance the training window. Two non-overlapping evaluation periods are used:

- **Window 1 (W1)**: Train January 2020–December 2022; Test January–December 2023 (post-bear-market recovery, BTC +154%).
- **Window 2 (W2)**: Train January 2021–December 2023; Test January–December 2024 (institutional-driven bull run, BTC +111%).

All hyperparameter tuning decisions used W1 exclusively. W2 is a fully held-out evaluation window; no tuning decisions were informed by its outcomes.

Figure 1 illustrates the two training and evaluation windows.

Fig. 1 Walk-forward evaluation protocol. Blue regions denote training periods; red regions denote held-out test periods. W1 tests on the 2023 recovery market; W2 tests on the 2024 bull run. The two evaluation windows cover qualitatively distinct market regimes, reducing the risk that results generalize only to a single regime type.

3.6 Exploratory Data Analysis

Prior to model development, the statistical properties of the three assets were examined to characterize return distributions and motivate metric selection.

Figure 2 presents the daily return distributions for all three assets. All distributions exhibit heavy tails and excess kurtosis (BTC: 11.1, ETH: 8.9, BNB: 34.7), confirming that extreme daily moves occur far more frequently than a normal distribution would predict. Standard deviation consequently underestimates true downside risk. This motivates penalizing only negative deviations (Sortino-style) rather than total variance (Sharpe-style).

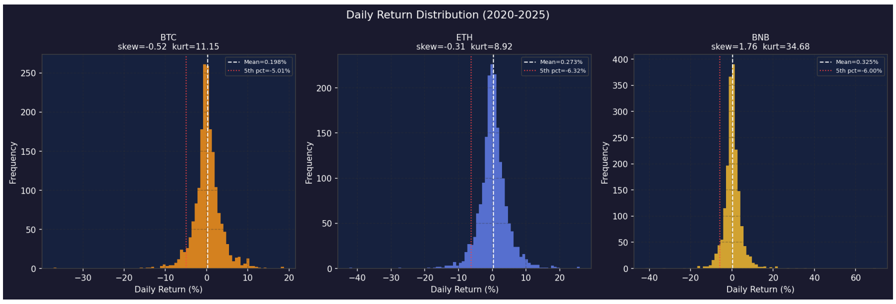

Fig. 2 Daily return distributions for BTC, ETH, and BNB (2020–2024). All three distributions exhibit heavy tails. BNB's extreme kurtosis (34.7) reflects infrequent but very large single-day moves, including the anomalous +69.8% event on 2021-02-19 that was excluded during cleaning.

Figure 3 shows 30-day rolling annualized volatility. Volatility is clearly time-varying and regime-dependent. The agent observes 14-day realized volatility as one of its six input features, providing a direct signal of current market conditions.

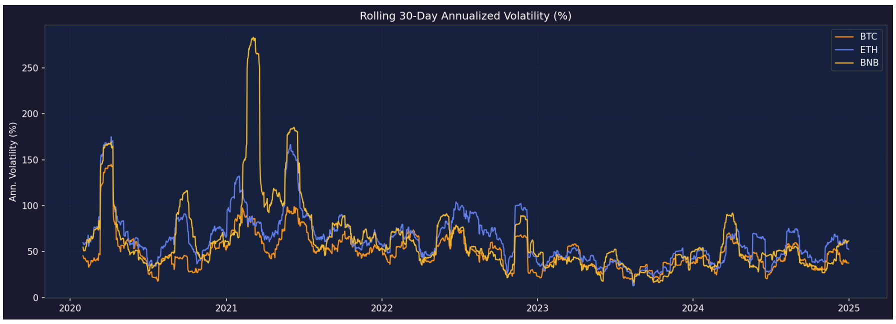

Fig. 3 Rolling 30-day annualized volatility for BTC, ETH, and BNB. Periods of low volatility (mid-2023) alternate with sharp spikes (early 2024). An agent that observes current volatility can learn to reduce exposure during high-volatility regimes.

Figure 4 presents pairwise return correlations. The three assets are highly correlated (BTC–ETH: 0.82, BTC–BNB: 0.66, ETH–BNB: 0.68), meaning diversification across them provides limited protection during systematic market-wide downturns.

This finding further motivates including cash as a genuinely uncorrelated defensive position.

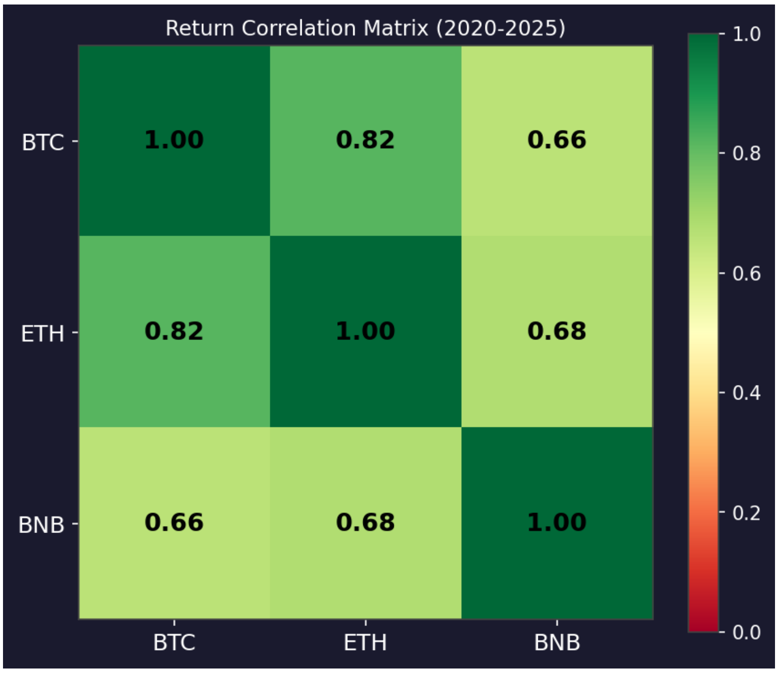

Fig. 4 Pairwise return correlations for BTC, ETH, and BNB (full period 2020–2024). High inter-asset correlations limit the diversification benefit of holding only cryptocurrency assets, strengthening the case for a dedicated cash allocation.

4 Methodology

Figure 5 shows the full pipeline. The subsections below describe each component.

4.1 MDP Formulation

Portfolio management is formulated as a Markov Decision Process. On each trading day, the agent observes state s_t, selects action a_t, receives reward r_t, and transitions to the next state. The objective is a policy $\pi_\theta(a_t|s_t)$ that maximizes expected discounted cumulative reward:

$$\mathbb{E}\left[\sum_{t=0}^{T}\gamma^t r_t\right] \tag{2}$$

4.1.1 State Space

The state vector contains 547 real-valued components organized into three groups:

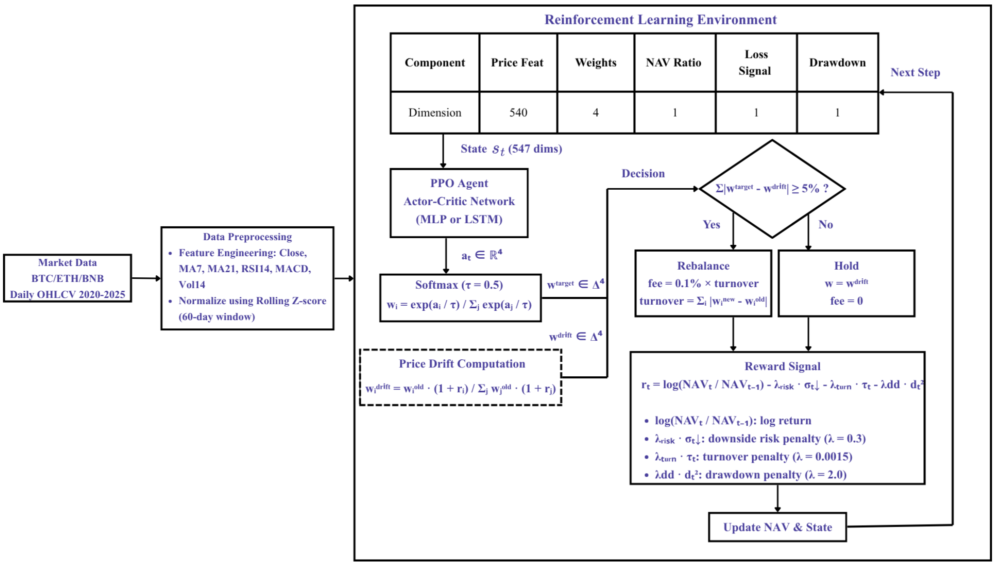

Fig. 5 System pipeline. Market data is processed through a feature engineering stage and normalized before being passed to the RL environment. Each trading day, the agent reads the state vector (30 days of market features, current portfolio weights, NAV ratio, loss streak, and drawdown level), outputs raw logits, and a temperature-scaled softmax converts them to target weights. The no-trade zone then compares the target against price-drifted weights; if the deviation exceeds 5%, the portfolio rebalances and fees are deducted. Otherwise the agent holds at zero cost. The reward combines log return, risk penalty, turnover penalty, and quadratic drawdown penalty.

$$s_t = \Big[\ \underbrace{X_{t-W+1:t}}_{\text{market context}}\ \Big\|\ \underbrace{w_t \,\|\, \rho_t}_{\text{portfolio status}}\ \Big\|\ \underbrace{\ell_t \,\|\, d_t}_{\text{risk signals}}\ \Big] \tag{3}$$

Market context (540 dimensions).

Thirty days of historical observations, three assets, six features each: $30 \times 3 \times 6 = 540$. This window provides sufficient context for the agent to identify short-term trends and recent volatility regimes.

Portfolio status (5 dimensions).

Current allocation weights for BTC, ETH, BNB, and cash (constrained to sum to 1), and the NAV ratio $\rho_t = \mathrm{NAV}_t/\mathrm{NAV}_0$. Without access to its current allocation, the agent cannot estimate the cost of reaching a new target. The NAV ratio conveys the agent's cumulative performance since inception.

Risk signals (2 dimensions).

A loss streak signal $\ell_t = 1 - e^{-0.5c_t}$, where c_t is the count of consecutive losing days, mapped to $[0, 1)$ via an exponential transform. A drawdown signal $d_t = \max(0, 1 - \mathrm{NAV}_t/\mathrm{NAV}_t^{\mathrm{peak}})$ measures the fraction by which the portfolio has declined from its historical peak. Both signals appear in the observation (enabling learned responses) and in the reward function (as penalties).

4.1.2 Action Space

The network outputs four unconstrained real values $a_t \in \mathbb{R}^4$. These are mapped to valid portfolio weights via temperature-scaled softmax:

$$w_{t,i}^{\text{target}} = \frac{\exp(a_{t,i}/\tau)}{\sum_{j=1}^{4} \exp(a_{t,j}/\tau)} \tag{4}$$

The temperature parameter τ controls the sharpness of the allocation distribution. For the same raw actions $a = [2.0, 1.5, 1.0, 0.5]$:

τ	BTC	ETH	BNB	Cash	Effect
1.0	34%	26%	20%	15%	Spread thin; concentrated positions infeasible
0.5	47%	26%	16%	9%	Decisive positions possible; not reckless
0.1	98%	2%	0%	0%	Near-total concentration

A temperature of $\tau = 0.5$ is used throughout. At $\tau = 1.0$, moderate logit gaps produce relatively spread allocations: as shown in the table above, a logit advantage of 0.5 per step yields only 34% for the top asset, limiting the agent's ability to act on strong directional signals. At $\tau = 0.1$, small logit differences produce near-binary allocations, which destabilizes learning. At $\tau = 0.5$, the agent retains the ability to shift a dominant fraction into cash during a downturn while still exploring meaningfully. This parameterization is consistent with Boltzmann-style action selection [9].

4.2 Cash Asset Definition

The fourth allocation target is a cash position, modeled as a zero-return, zero-volatility asset equivalent to holding USDT (Tether) or an equivalent stablecoin. It earns no interest and incurs no holding cost. Transactions into or out of cash are subject to the same 0.1% exchange fee as any other rebalancing trade. This formulation makes cash a genuine uncorrelated refuge: when the agent shifts weight from crypto to cash, portfolio variance decreases but the rebalancing trade itself still carries a fee. The agent must therefore learn that the protective benefit of holding cash outweighs the round-trip transaction cost.

4.3 Transaction Cost Model

A fee rate of $\phi = 0.001$ (Binance spot fee) is applied to realized turnover on each rebalancing event:

$$\text{cost} = \phi \cdot \text{turnover}, \quad \text{turnover} = \sum_{i=1}^{4} |w_i^{\text{new}} - w_i^{\text{old}}| \tag{5}$$

Both buy-side and sell-side transactions contribute to turnover. As cryptocurrency markets operate continuously (365 days per year), a portfolio sustaining 10% daily turnover incurs approximately $0.001 \times 0.10 \times 365 \approx 3.65\%$ per year in fees alone.

4.4 The No-Trade Zone

Natural price movements cause the portfolio composition to drift from the previous allocation continuously. A rebalancing rule that triggers on any deviation from the target will therefore trade on every day that prices move, regardless of whether the trade is cost-justified.

The proposed mechanism first computes drifted weights, defined as the portfolio composition after today's price movements, before any agent action:

$$w_i^{\text{drift}} = \frac{w_i^{\text{old}} \cdot (1 + r_{t,i})}{\sum_{j=1}^{4} w_j^{\text{old}} \cdot (1 + r_{t,j})} \tag{6}$$

Rebalancing is then conditioned on the deviation between the target and the drifted weights:

$$\text{rebalance} = \begin{cases} \text{Yes,} & \text{if } \sum_i |w_i^{\text{target}} - w_i^{\text{drift}}| \geq \theta \\ \text{No,} & \text{otherwise; hold at zero cost} \end{cases} \tag{7}$$

The threshold $\theta = 0.05$ (5%) was selected to match the typical daily drift magnitude of the three-asset portfolio. BTC, ETH, and BNB individually swing 2–3% per day; combined portfolio drift is approximately 1–2% daily. A threshold below 3% permits near-daily trading; a threshold above 10% allows excessive drift from the agent's intended allocation. At 5%, a trade is triggered only when at least 2.5% of capital genuinely warrants redistribution. Betancourt and Chen [2] reached an analogous conclusion: small weight adjustments yield negligible net return improvements once fees are accounted for.

4.5 Reward Function

The reward function comprises four terms, each targeting a specific failure mode observed in preliminary experiments:

$$r_t = \underbrace{\log \frac{\text{NAV}_t}{\text{NAV}_{t-1}}}_{\text{growth}} - \underbrace{\lambda_{\text{risk}} \cdot \sigma_t^{\downarrow}}_{\text{risk penalty}} - \underbrace{\lambda_{\text{turn}} \cdot \tau_t}_{\text{turnover penalty}} - \underbrace{\lambda_{\text{dd}} \cdot d_t^2}_{\text{drawdown penalty}} \tag{8}$$

4.5.1 Log Return

The growth term is the log ratio of consecutive NAV values. Log returns are additive over time, aligning naturally with the cumulative reward objective of the RL framework.

4.5.2 Risk Penalty (lambda_risk = 0.3)

Three variants are evaluated in Experiment 1:

- **Sortino-style**: penalizes only downside volatility over a 14-day rolling window. Positive deviations are not penalized.
- **Sharpe-style**: penalizes total volatility, treating upside and downside deviations symmetrically.
- **Raw**: no risk penalty; included as a control condition.

The coefficient $\lambda_{\text{risk}} = 0.3$ was selected via grid search. Typical daily log returns fall in the range 0.003–0.01. Downside deviation during volatile regimes runs 0.005–0.02. At $\lambda_{\text{risk}} = 0.3$, the penalty is negligible in calm markets and materially negative during high-volatility periods.

It is important to note that the Sortino-style reward variant is related to, but not equivalent to, the Sortino ratio used as an evaluation metric. The reward term penalizes rolling 14-day downside deviation at each step, shaping the agent's behavior during training. The evaluation Sortino ratio is computed post-hoc from the full NAV series using standard annualization. The two quantities share the same conceptual motivation (penalizing downside risk) but differ in window, normalization, and purpose.

4.5.3 Turnover Penalty (lambda_turn = 0.0015)

The no-trade zone provides a binary gate on whether a trade occurs. The turnover penalty provides a graduated incentive on trade size, discouraging large-scale reshuffles when the gate permits trading. The two mechanisms are complementary: the gate controls trade frequency; the penalty modulates trade magnitude. Importantly, this penalty does not double-count the transaction cost already deducted from NAV. The NAV deduction represents the actual financial cost; the turnover penalty is a reward-shaping term that creates a learning gradient toward smaller trades, improving training stability in regimes where the no-trade threshold has been cleared.

4.5.4 Drawdown Penalty (lambda_dd = 2.0)

The penalty is proportional to the square of the current drawdown fraction. This formulation reflects the asymmetry of financial loss recovery [3]: recovering a 10% loss requires an 11.1% gain; recovering a 20% loss requires 25%; recovering a 40% loss requires 66.7%. The quadratic scaling ensures that small drawdowns generate only background noise while deep losses create dominant penalty signals:

Drawdown	d_t^2	Penalty ($\lambda_{\text{dd}} = 2.0$)	vs. daily return
5%	0.0025	0.005	Negligible
10%	0.01	0.020	Noticeable
20%	0.04	0.080	Dominant over typical growth signal
30%	0.09	0.180	Agent incentivized to reduce exposure

At 30% drawdown, the penalty exceeds the typical daily growth signal by an order of magnitude, creating a strong incentive to shift capital toward cash without any explicit sell rule.

4.5.5 Reward Clipping

The final reward is clamped to $[-5, +5]$. Cryptocurrency markets can produce extreme single-day moves; without clipping, outliers generate large gradient updates that destabilize training.

4.6 PPO: The Learning Algorithm

Proximal Policy Optimization [10] is used throughout this study. PPO handles continuous action spaces and is well-established across a wide range of financial environments. Two networks form the actor-critic architecture:

- The **Actor** (π_θ): maps state observations to a distribution over allocation decisions.
- The **Critic** (V_ϕ): maps state observations to a scalar estimate of expected future discounted reward.

The agent collects 2,048 environment steps (approximately eight simulated months) before each parameter update.

4.6.1 Actor Update

The clipped surrogate objective prevents destabilizing policy updates:

$$L^{\text{clip}}(\theta) = \mathbb{E}_t \left[\min\left(\frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)} \hat{A}_t, \;\; \text{clip}\left(\frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\text{old}}}(a_t|s_t)}, 0.8, 1.2 \right) \hat{A}_t \right) \right] \tag{9}$$

The probability ratio $\pi_\theta/\pi_{\theta_{\text{old}}}$ is clipped to $[0.8, 1.2]$, limiting the magnitude of any single update step. The advantage estimate $\hat{A}_t$ indicates whether a given action produced better or worse outcomes than the critic's expectation.

4.6.2 Critic Update

$$L^V(\phi) = \mathbb{E}_t \left[\left(V_\phi(s_t) - V_t^{\text{target}} \right)^2 \right] \tag{10}$$

The advantage signal that guides actor updates is computed from the critic's value estimates. Inaccurate value estimation in noisy financial environments can propagate errors into policy gradients, causing training instability.

4.6.3 Combined Loss

$$L(\theta, \phi) = -L^{\text{clip}}(\theta) + 0.7 \cdot L^V(\phi) - 0.03 \cdot H[\pi_\theta] \tag{11}$$

The critic coefficient is set to 0.7 (rather than the standard 0.5) to prioritize value function accuracy under noisy financial reward signals. The entropy bonus $H[\pi_\theta]$ prevents premature policy collapse, keeping the agent exploring during the highly variable market regimes in the training data.

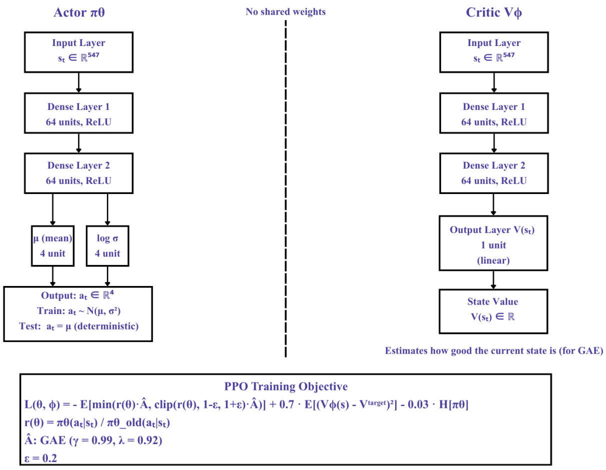

Fig. 6 Actor-Critic architecture. Left (Actor): 547-dimensional state input, two hidden layers of 64 units with ReLU activation, output layer producing μ and $\log \sigma$ (4 values each). During training, actions are sampled as $a \sim \mathcal{N}(\mu, \sigma^2)$; at evaluation, μ is used deterministically. Softmax ($\tau = 0.5$) converts actions to portfolio weights. Right (Critic): identical structure, single scalar output $V(s_t)$. No weights are shared between actor and critic.

4.6.4 Generalized Advantage Estimation

$$\delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t) \tag{12}$$

$$\hat{A}_t = \delta_t + (\gamma \lambda_{\text{GAE}})\delta_{t+1} + (\gamma \lambda_{\text{GAE}})^2 \delta_{t+2} + \cdots \tag{13}$$

With $\lambda_{\text{GAE}} = 0.92$, the contribution of a TD error l steps ahead falls below 5% after approximately 32 steps. In practice, each allocation decision is evaluated primarily against outcomes over the following month.

The discount factor $\gamma = 0.99$ was selected via grid search. At $\gamma = 0.97$, agents displayed impatient behavior, liquidating during temporary drawdowns and forgoing subsequent recoveries. At $\gamma = 0.99$, a reward 100 days ahead retains 37% of its face value, providing sufficient incentive for patient capital preservation.

4.7 Network Architecture

4.7.1 MLP

The standard feedforward architecture (Figure 6) takes the 547-dimensional state as input. Two hidden layers of 64 neurons each use ReLU activation. The actor produces eight outputs (four for μ, four for $\log \sigma$); the critic produces a single scalar. No weights are shared between the two networks.

4.7.2 LSTM

The first hidden layer is replaced by an LSTM of 256 units, implemented via RecurrentPPO from the sb3-contrib library, with batch size reduced to 128 to accommodate sequential data processing. The MLP processes each step independently using a fixed

30-day feature window. The LSTM maintains a hidden state across the full episode, enabling the detection of multi-week patterns that would not be fully captured within a 30-day window alone; for example, a three-week gradual decline that the MLP observes only through lagged indicators.

4.8 Hyperparameter Tuning

Grid search was conducted over 54 combinations using a nested split within Window 1 only. Specifically, the W1 training period (January 2020–December 2022) was further divided: January 2020–June 2022 served as the grid search training set, and July–December 2022 served as the internal validation set. The W1 test period (2023) was not touched at any stage of hyperparameter selection; it was reserved exclusively for final evaluation. Window 2 was not used in any way during tuning. This protocol ensures that no hyperparameter decision was informed by held-out test data.

Table 1 Hyperparameter search space and selected values. Best combination selected by highest Sortino ratio on the W1 internal validation set (Jul–Dec 2022), which is strictly within the W1 training window.

Parameter	Candidates	Selected
Learning rate	$\{10^{-5},\ 3 \times 10^{-5}\}$	10^{-5}
Discount γ	$\{0.97,\ 0.985,\ 0.99\}$	0.99
Entropy c_H	$\{0.01,\ 0.02,\ 0.03\}$	0.03
Risk coefficient λ_{risk}	$\{0.3,\ 0.5,\ 1.0\}$	0.3

The rationale for each selected value:

- **$\mathbf{10^{-5}}$ learning rate**: The larger rate (3×10^{-5}) produced training curves with periodic collapses consistent with step-size instability under noisy cryptocurrency rewards.
- **$\boldsymbol{\gamma} = \mathbf{0.99}$**: Agents with $\gamma = 0.97$ exhibited excessive short-horizon optimization, liquidating during temporary pullbacks and missing subsequent recoveries.
- **$\mathbf{c_H = 0.03}$**: Higher entropy maintained agent flexibility across market regimes. Lower values caused premature specialization that degraded performance during regime transitions.
- **$\boldsymbol{\lambda}_{\text{risk}} = \mathbf{0.3}$**: Larger values (0.5 and 1.0) produced excessively conservative agents that held cash throughout clear bull markets, forgoing substantial return.

Fixed parameters not subject to grid search: $n_{\text{steps}} = 2048$, batch size $= 256$, $\lambda_{\text{GAE}} = 0.92$, critic coefficient $c_V = 0.7$, clip range $\epsilon = 0.2$ [10]. Grid search runs used 150K training steps. Final training ran 500K steps with early stopping (halt if validation reward stalls for 8 consecutive evaluations, with a minimum of 6 evaluations before early stopping activates).

4.9 Training Procedure

One environment step corresponds to one calendar day. Each episode spans 365 steps, reflecting the continuous 24/7 nature of cryptocurrency markets. Unlike traditional equity markets, crypto trades every day of the year with no exchange holidays, making 365 the appropriate episode length. The daily execution loop:

1. Price and indicator data advance to the current step.
2. The agent reads the state vector and produces a raw action.
3. Temperature-scaled softmax converts the action to target weights (Eq. 4).
4. The no-trade zone evaluates the deviation from drifted weights (Eq. 7).
5. If rebalancing: transaction fees are deducted from NAV. If holding: weights drift naturally at zero cost.
6. The reward is computed (Eq. 8) and clipped to $[-5, +5]$.
7. Time advances to the next step.

Three models per configuration are trained using seeds 42, 123, and 777. Episode start indices are drawn uniformly within the training window, ensuring that the agent encounters bear markets, bull runs, and sideways periods throughout training.

5 Experiments

Three experiments share the hyperparameter configuration from Table 1.

5.1 Exp 1: Reward Function Comparison

MLP agents are trained with each of the three reward variants (Sortino, Sharpe, Raw) and evaluated on both walk-forward windows. The primary question is whether the choice of risk metric materially affects agent behavior, or whether the no-trade zone dominates the cost profile regardless of reward formulation.

5.2 Exp 2: PPO vs. Passive Baselines

The best PPO variant (Sortino reward, mean over three seeds) is compared against three passive benchmarks:

- **Equal-Weight (EW)**: Initial allocation of 33.3% each in BTC, ETH, and BNB. Weights drift naturally with prices and are never rebalanced; no trades, no fees. This is a buy-and-hold equal-weight strategy, not a daily-rebalanced one. Periodic rebalancing of equal weights would incur substantial transaction costs and would constitute an unfair upper bound on passive performance.
- **Buy & Hold BTC (B&H)**: 100% Bitcoin from inception. Zero trades. Over most multi-year windows in this dataset, simply holding Bitcoin outperforms most active strategies once fees are included.
- **Market-Cap Weighted (MCW)**: Fixed allocations proportional to approximate market capitalization: BTC 60%, ETH 30%, BNB 10%. Passive hold, no rebalancing, no fees.

All three baselines incur zero transaction costs by construction, which provides a structural advantage over the PPO agent. Any risk-adjusted performance advantage observed for PPO is therefore achieved despite this cost asymmetry.

5.3 Exp 3: MLP vs. LSTM

Both architectures are evaluated with the Sortino reward across three seeds and both windows. The experiment examines whether the LSTM's episodic memory provides a meaningful advantage over the MLP's fixed 30-day feature window in the context of daily cryptocurrency trading.

5.4 Metrics

All metrics are computed on held-out test data using a deterministic policy (no exploration noise):

- **Total Return**: end-to-end portfolio growth (%).
- **Sharpe Ratio**: $\bar{r}/\mathrm{std}(r) \times \sqrt{365}$ (annualization factor of 365 reflects the continuous 24/7 trading calendar of cryptocurrency markets).
- **Sortino Ratio**: as Sharpe but with downside-only volatility in the denominator.
- **Max Drawdown**: worst peak-to-trough decline (%). Lower is better.
- **Calmar Ratio**: annualized return divided by max drawdown.
- **Trading Frequency**: fraction of days on which a rebalancing trade was executed (%).
- **Total Fees**: cumulative transaction costs as a fraction of starting capital (%).

6 Results and Discussion

6.1 Exp 1: Reward Function Comparison

Table 2 reports performance averaged across W1 and W2, with each PPO result representing the mean over three seeds.

Table 2 Exp 1: Reward function comparison (average over W1 and W2, mean across seeds 42/123/777). Bold indicates best value among the three PPO variants.

Method	Return (%)	Sortino	MDD (%)
PPO-Sortino	**54.3**	**1.627**	22.0
PPO-Sharpe	53.7	1.613	22.0
PPO-Raw	53.6	1.613	22.0

All three reward variants produced nearly identical results. Return differences are below 1%, Sortino ratios differ by less than 0.015, and maximum drawdowns are identical across the three agents. Minor differences in trading frequency are visible: PPO-Sortino traded on 13.3% of days versus 14.5% for PPO-Sharpe. All three agents achieved annual fees below 0.6%, substantially lower than the unconstrained baseline system prior to introducing the no-trade zone. The primary finding of this experiment is not the superiority of any reward variant; it is that the no-trade zone dominates the cost profile regardless of which risk metric is optimized.

6.2 Exp 2: PPO vs. Passive Baselines

Table 3 compares PPO-Sortino against the three passive baselines, averaged over both evaluation windows.

Table 3 Exp 2: PPO-Sortino vs. passive baselines (average over W1 and W2). Bold indicates the lowest MDD. All passive baselines incur zero transaction costs by construction.

Method	Return (%)	Sortino	MDD (%)
PPO-Sortino	54.3	1.627	**22.0**
Equal-Weight	85.9	1.676	30.8
B&H BTC	132.9	2.401	23.1
MktCap-Weight	103.0	1.972	27.3

PPO-Sortino underperforms all three baselines on raw total return. The primary explanation is the agent's maintained cash allocation of approximately 25%, which reduced market exposure during the 2023–2024 Bitcoin bull run. An agent designed for capital preservation will underperform a fully-invested passive position during sustained rallies; this is the intended performance trade-off.

The agent achieved the lowest maximum drawdown among all strategies at 22.0%, versus 30.8% for Equal-Weight and 27.3% for Market-Cap Weighted. This gap is most pronounced in Window 2 (2024), where PPO's Sortino ratio of 1.923 exceeded both Equal-Weight (1.534) and Market-Cap Weighted (1.541). This confirms that the risk management properties are meaningful and not merely artifacts of the particular test period.

6.3 Exp 3: MLP vs. LSTM

Table 4 compares the two architectures using Sortino reward, averaged over both windows and three seeds.

Performance differences between the two architectures are small. The LSTM achieves a marginally higher Sortino ratio (1.655 vs. 1.627), while the MLP delivers slightly higher total return, lower drawdown, and lower fees. The LSTM traded on 14.9% of days versus 13.3% for the MLP, which accounts for the fee and drawdown difference.

Results diverged across evaluation windows. In W1 (2023), LSTM averaged a 36.6% return versus MLP's 35.8%, suggesting some benefit from episodic memory in the recovery-phase market. In W2 (2024), the relationship reversed: MLP returned 72.8% while LSTM returned 69.9%. Neither architecture consistently dominated.

Table 4 Exp 3: MLP vs. LSTM policy (Sortino reward, average over W1 and W2, mean across seeds). Bold indicates best value between the two PPO variants. Passive baselines are shown as reference only.

Method	Return (%)	Sortino	MDD (%)
PPO-MLP	**54.3**	1.627	**22.0**
PPO-LSTM	53.2	**1.655**	22.8
Equal-Weight	85.9	1.676	30.8
B&H BTC	132.9	2.401	23.1

Training costs differ substantially. On an Intel Core i7-13700K (13th Gen), the MLP configuration required approximately 2–3 hours per full training run (500K steps, three seeds, both windows). The LSTM configuration required approximately 12 hours under equivalent conditions, reflecting the sequential nature of LSTM computation and the reduced batch size of 128. Given that the LSTM provides no consistent performance advantage, the MLP is the preferred architecture for this task on computational efficiency grounds.

Overall, the 30-day market context window in the state representation appears to capture the most relevant temporal patterns for daily trading at this frequency. The additional expressiveness of the LSTM introduces higher variance between seeds without a commensurate performance benefit.

7 Live Evaluation: March 2025 – March 2026

To assess out-of-sample robustness beyond the two training windows, the best PPO-MLP model (Sortino reward, W2 training) was evaluated on live market data from March 1, 2025 to March 23, 2026, a period entirely outside any training or tuning decision. This period coincides with a broad cryptocurrency market downturn driven by macroeconomic uncertainty, with Bitcoin declining approximately 19% from its early-March 2025 levels.

7.1 Single-Agent Backtest

Figure 7 shows the NAV curve and daily portfolio weight allocations over the evaluation period.

The agent recorded a total return of −2.88% with total fees of 0.26%, closing with an allocation of approximately 30% CASH, 25% BNB, 25% BTC, and 20% ETH. The maintained cash position reflects the agent's learned defensive behavior: under sustained negative returns and elevated drawdown signals in the state vector, capital was gradually shifted away from risky assets without any hard-coded exit rule. Maximum drawdown over the period reached −44.5% on an intra-period rolling basis, consistent with the severity of the broader market conditions.

Fig. 7 Live backtest on March 1, 2025 – March 23, 2026. Top: NAV curve of the PPO agent versus individual asset price series. Bottom: daily portfolio weight allocation across CASH, BNB, BTC, and ETH. The agent maintained approximately 30% cash throughout, providing a partial buffer against the broader market decline.

7.2 Strategy Comparison

Figure 8 compares the PPO agent against seven alternative strategies over the same period.

Table 5 summarizes key metrics for all eight strategies.

Table 5 Live evaluation results, March 1, 2025 – March 23, 2026. All strategies initialized at $10,000.

Strategy	Return (%)	Sharpe	Sortino	Max DD (%)
Momentum	−2.78	−0.039	−0.052	−53.91
RL Model 1 (ours)	**−3.16**	**−0.062**	**−0.082**	**−43.86**
60/40 Portfolio	−3.72	−0.091	−0.119	−37.56
Equal-Weight	−6.12	−0.091	−0.119	−54.39
Market-Cap Weight	−12.01	−0.189	−0.252	−53.36
RSI-Based	−16.29	−0.260	−0.339	−54.85
Buy & Hold BTC	−18.42	−0.344	−0.462	−49.53
Min Variance	−19.78	−0.370	−0.465	−49.92

Chiến lược	Return (%)	Sharpe	Sortino	Max DD (%)
Momentum	-2.78%	-0.039	-0.052	-53.91%
RL Model 1	-3.16%	-0.062	-0.082	-43.86%
60/40 Portfolio	-3.72%	-0.091	-0.119	-37.56%
Equal Weight	-6.12%	-0.091	-0.119	-54.39%
Market-Cap Weight	-12.01%	-0.189	-0.252	-53.36%
RSI-Based	-16.29%	-0.260	-0.339	-54.85%
Buy & Hold BTC	-18.42%	-0.344	-0.462	-49.53%
Min Variance	-19.78%	-0.370	-0.465	-49.92%

Fig. 8 Strategy comparison on March 1, 2025 – March 23, 2026. NAV curves (top) and summary metrics (bottom) for eight strategies: Equal-Weight, Buy & Hold BTC, Market-Cap Weight, Momentum, Min Variance, 60/40 Portfolio, RSI-Based, and RL Model 1 (the proposed PPO agent). All strategies initialized at $10,000.

The PPO agent ranked second among eight strategies with a return of -1.16%, outperforming all purely crypto-allocated passive strategies. Only Momentum, a rule-based strategy that rotates into the best-performing prior-period asset, produced a smaller loss (-1.78%). The 60/40 Portfolio (60% crypto, 40% stablecoin) achieved lower maximum drawdown (-37.56%) but a worse return (-3.72%), indicating that the agent's learned cash allocation was more dynamically responsive to market conditions than a fixed 40% defensive position.

These results carry a significant caveat. The evaluation window spans approximately one year, but encompasses a specific bear-market episode at its onset; a single market regime is insufficient to establish general robustness. Nevertheless, the results are consistent with the in-sample experimental findings: the primary value of the proposed agent lies in limiting downside exposure rather than maximizing returns, and this characteristic appears to generalize to an entirely unseen market regime.

8 Limitations

1. **Commission fees only.** The cost model does not include bid-ask spreads or market impact. For larger positions or thinner order books, these implicit costs could increase total execution cost by 1.5–2×.
2. **Daily resolution only.** Intraday agents could respond more quickly to market conditions but would face more frequent fee events and require substantially more training data to learn stable policies.
3. **Three assets.** Extending to 50 or 100 tokens would expand the action space significantly. The softmax projection may remain valid, but training complexity and data requirements would grow substantially.
4. **Execution at close price.** Market orders execute at prices that typically differ from the quoted close. In a continuous 24/7 market, the definition of a closing price also varies by data provider.
5. **Market regime evolution.** Regulatory changes, exchange events, protocol forks, and macroeconomic shocks can render a trained policy suboptimal. Walk-forward validation reduces this risk, but periodic retraining would be required for any real-world deployment.

9 Conclusion

This study presented a transaction-cost-aware deep reinforcement learning framework for cryptocurrency portfolio management. The system addresses a specific and practically significant failure mode: agents that generate nominal alpha while simultaneously destroying it through excessive trading fees. Three components work in combination: a price-drift-adjusted no-trade zone that suppresses fee-negative rebalancing, a cash allocation slot that the agent learns to use defensively, and a quadratic drawdown penalty that creates escalating incentives for capital preservation.

Evaluated on five years of BTC, ETH, and BNB data, the no-trade zone alone substantially reduced annual transaction costs, to below 0.6% in all experimental configurations. Across three reward formulations the performance differences were negligible, pointing to a clear conclusion: the cost-control mechanism is the dominant contributor to the system's practical effectiveness, not the specific risk metric embedded in the reward function. The cash mechanism functioned as intended: capital was reallocated toward the defensive position during drawdown periods without any hard-coded trigger rules.

The MLP and LSTM architectures produced comparable results across both evaluation windows. The LSTM's episodic memory provided no consistent advantage over the MLP's 30-day feature window at daily trading frequency, while requiring approximately 4× the wall-clock training time. Out-of-sample evaluation over the March 2025 – March 2026 period further confirmed the agent's capacity to limit downside exposure in an unseen market regime, ranking second among eight strategies during the initial bear-market phase.

Future directions include incorporating realistic spread and slippage models, expanding the asset universe, and developing mechanisms to maintain policy currency under non-stationary market regimes.

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Declarations

- **Funding:** Not applicable.
- **Conflict of interest:** The authors declare no conflict of interest.
- **Data availability:** All price data were obtained from Yahoo Finance via the `yfinance` Python library. The data are publicly accessible.
- **Code availability:** Code is available upon reasonable request.
- **Author contribution:** Nguyen Trong Hieu designed the reward function, implemented the RL environment, and conducted the hyperparameter grid search. Luu Duc Anh implemented the PPO and RecurrentPPO training pipeline and ran all training experiments. Nguyen Duc Hien built the evaluation framework, backtest infrastructure, and baseline comparisons. All authors contributed to writing and reviewing the manuscript.

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